

Present a Complete Design Walkthrough of an EER Diagram for a Retail Chain Management System

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What We'll Cover

01

EER Fundamentals

Concepts that extend the classic ER model
— specialization, generalization, constraints,
categories.

Slides 03–07

02

Retail Chain Management System

The business domain: stores, staff, products,
suppliers and customers.

Slides 08–10

03

EER Diagram Walkthrough

Building the diagram piece by piece, then
viewing it as one complete design.

Slides 11–16

What Is an Extended ER (EER) Model?

The Enhanced Entity-Relationship (EER) Model builds on the classic ER Model by adding semantic constructs needed to represent **complex, real-world data** — such as objects that naturally fall into overlapping categories or role-based hierarchies.

- **Subclasses & Superclasses**

Group entities that share attributes, but also have their own distinct ones.

- **Specialization / Generalization**

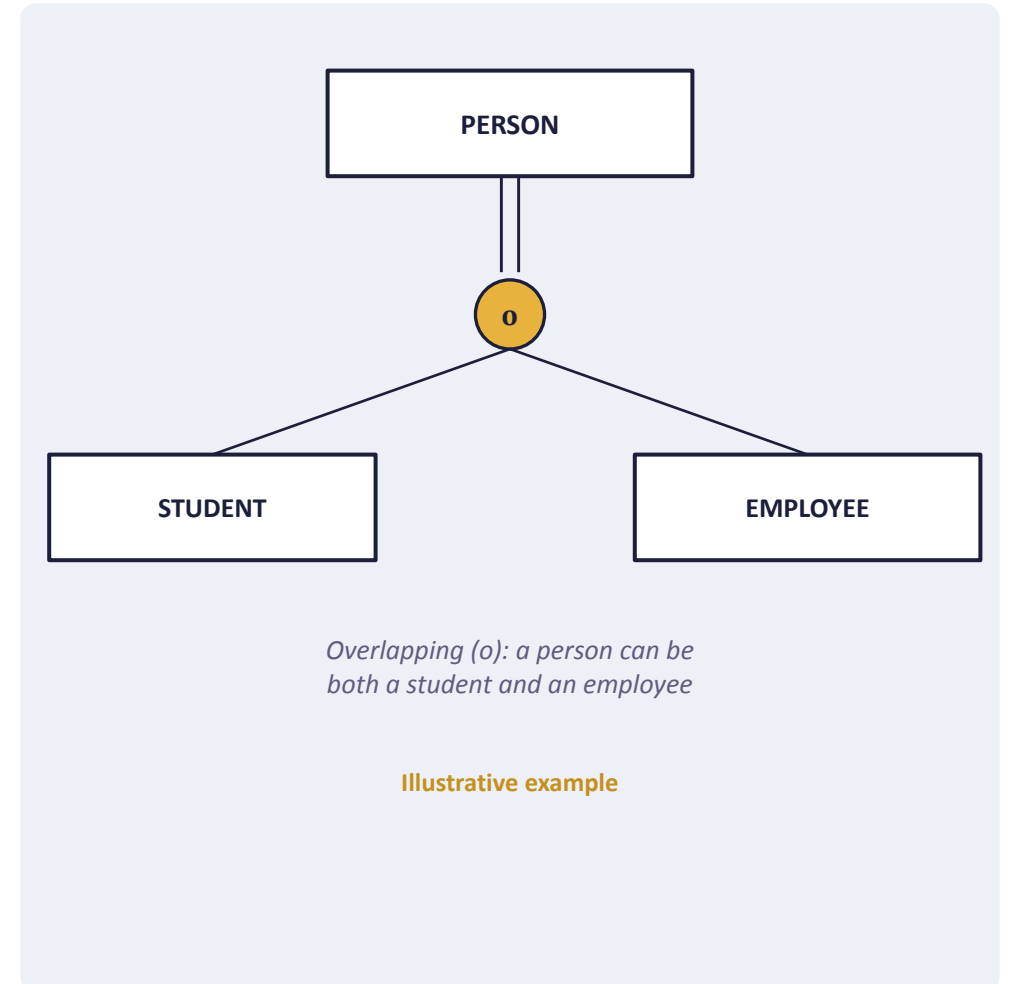
Top-down and bottom-up techniques for building class hierarchies.

- **Constraints**

Disjoint / overlapping and total / partial participation rules.

- **Inheritance**

Subclasses inherit every attribute and relationship of their superclass.



From ER to EER — Why We Need More

A plain ER diagram models entities, attributes and relationships well — but retail data has natural hierarchies and shared roles that ER alone cannot express cleanly.

BASIC ER MODEL

- Entities, attributes, relationships & keys
- No way to model role hierarchies (e.g. Manager vs Cashier as kinds of Employee)
- Cannot express that Store subtypes share core attributes but differ in others
- Every entity treated as a flat, independent structure

ENHANCED ER (EER) MODEL

- ✓ Everything ER offers, plus subclass / superclass structures
- ✓ Specialization models Store → Flagship / Franchise / Outlet cleanly
- ✓ Generalization models Manager, Cashier, Warehouse Staff as kinds of Employee
- ✓ Constraints (disjoint/overlap, total/partial) capture real business rules precisely

Specialization & Generalization

SPECIALIZATION (Top-Down)

Start with a superclass and divide it into subclasses based on a distinguishing characteristic.

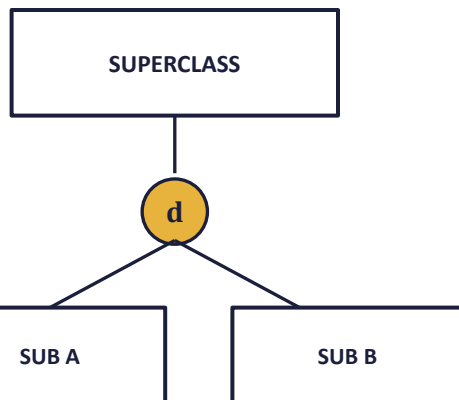
Example: EMPLOYEE is divided into MANAGER, CASHIER and WAREHOUSE STAFF because each performs a different job role.

GENERALIZATION (Bottom-Up)

Start with several similar subclasses and combine their shared attributes into one superclass.

Example: FLAGSHIP STORE, FRANCHISE STORE and OUTLET STORE are generalized into one STORE entity that holds their common attributes.

EER NOTATION USED IN THIS DECK



Circle = specialization/generalization
"d" = disjoint · "o" = overlapping



Double outline = weak entity
Double diamond = identifying relationship

Entities Identified for the EER Diagram

Analysing the components on the previous slide, we identify the following entity types before drawing the diagram:

ENTITY	TYPE	DESCRIPTION
CHAIN	Strong	The single retail brand / head office that owns all stores.
STORE	Superclass	Any retail outlet; specialized into Flagship, Franchise, Outlet.
EMPLOYEE	Superclass	Any staff member; specialized into Manager, Cashier, Warehouse Staff.
PRODUCT	Strong	An item sold across the chain, grouped into categories.
SUPPLIER	Strong	External vendor that supplies products to the chain.
INVENTORY	Weak	Stock level of a product at a specific store; identified by Store.
CUSTOMER	Strong	A shopper who makes purchases at any store.
SALES TRANSACTION	Strong	A single billed sale, linked to a store, cashier and customer.

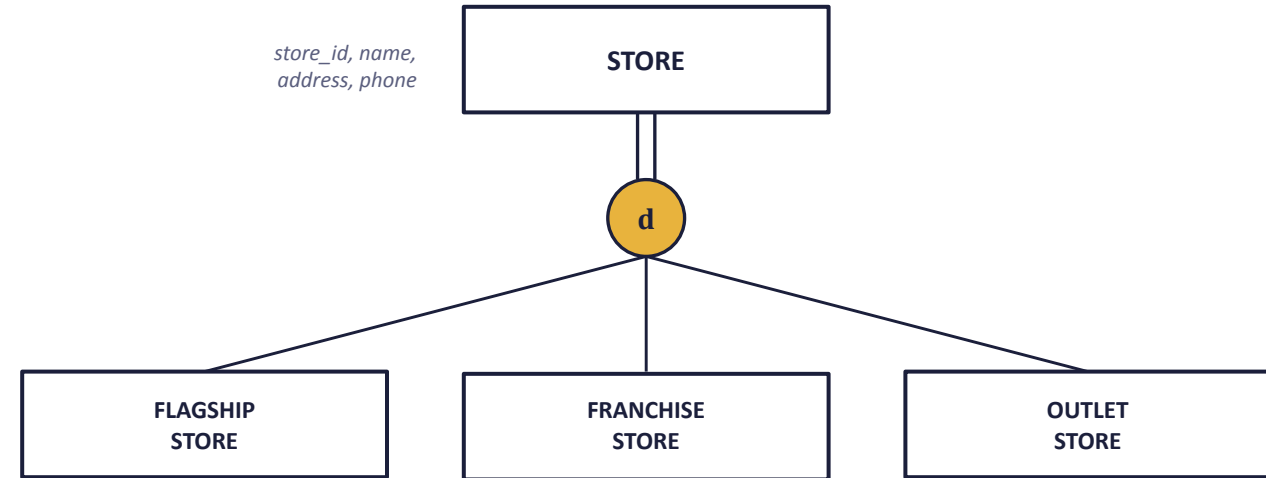
Building Block 1 — Store Specialization

STORE is generalized from three subtypes that share core attributes (store_id, address, phone) but differ in how they operate.

FLAGSHIP STORE

FRANCHISE STORE

OUTLET STORE



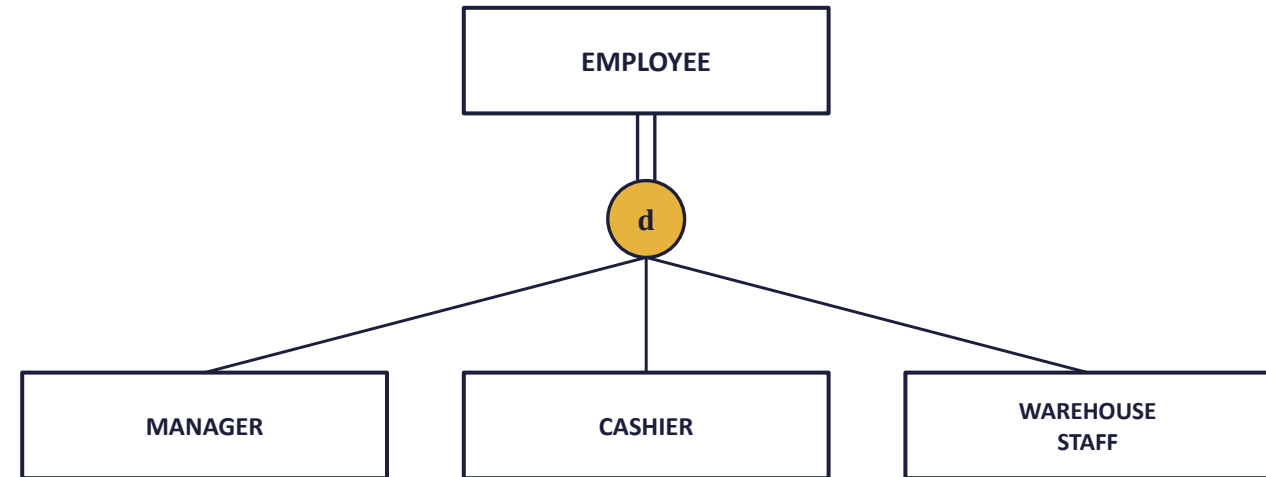
Building Block 2 — Employee Specialization

EMPLOYEE is specialized by job role. Since a staff member holds exactly one role at a time, the constraint is disjoint, total (d).

MANAGER

CASHIER

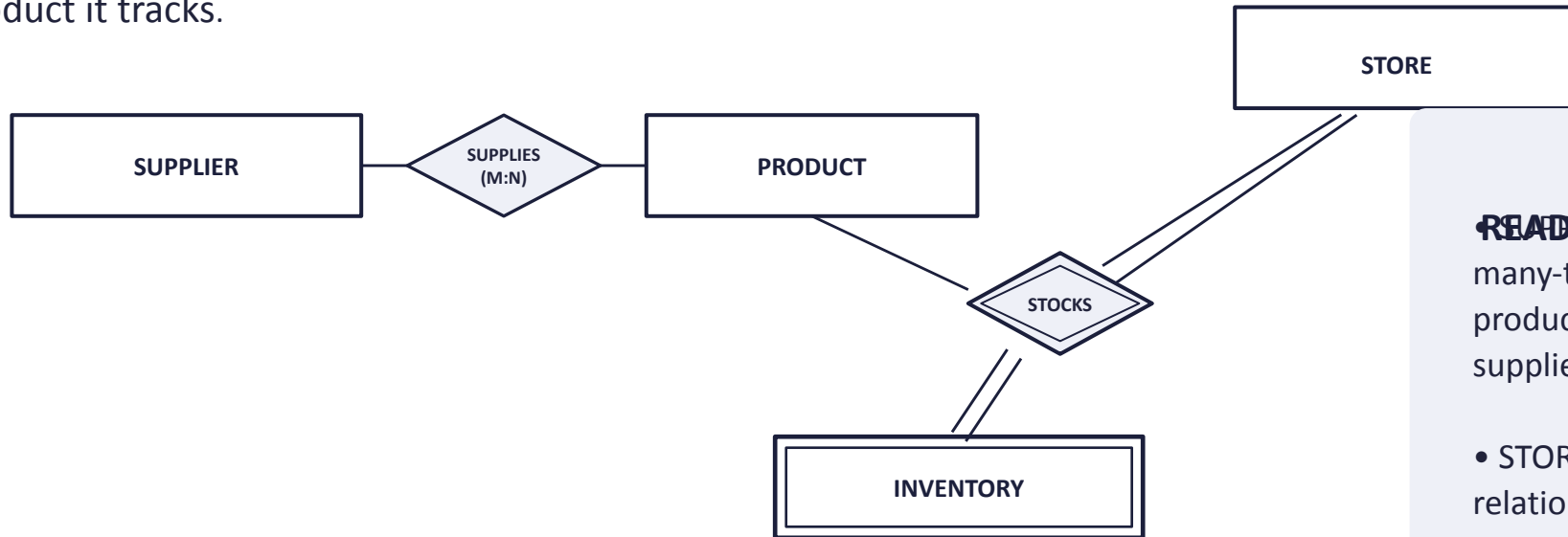
WAREHOUSE STAFF



CASHIER is the role that later connects to SALES TRANSACTION — only a cashier can process a sale, which the diagram enforces through this subclass rather than the generic EMPLOYEE entity.

Building Block 3 — Product, Inventory & Supplier

INVENTORY is a weak entity: a stock record only makes sense in the context of a specific STORE, and its partial key is the product it tracks.



READING THIS BLOCK

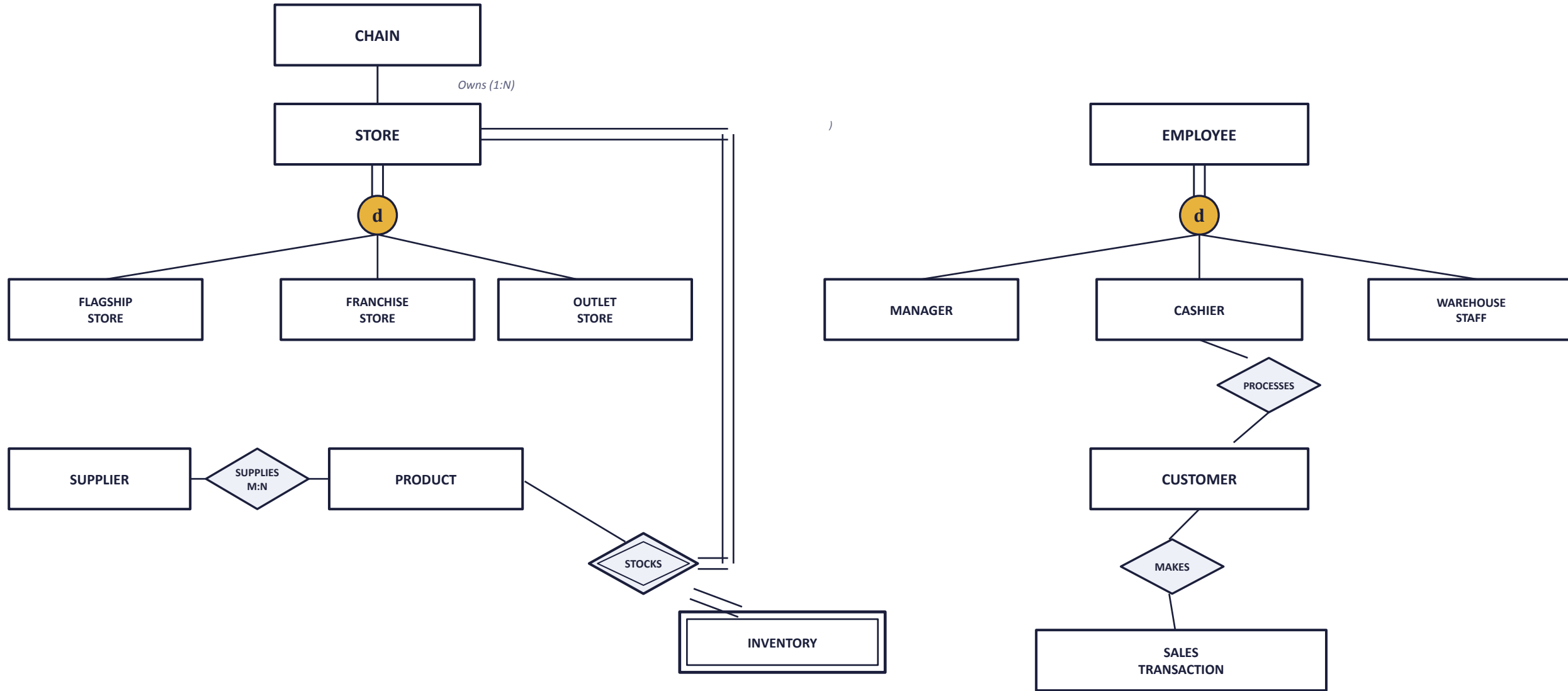
many-to-many: one supplier ships many products, one product may have several suppliers.

- STORE ↔ INVENTORY is the identifying relationship (double line/diamond) — an inventory row cannot exist without its store.
- PRODUCT feeds INVENTORY with the item being tracked, but is not the identifying owner.

Relationship Summary

RELATIONSHIP	CONNECTS	CARDINALITY
Owns	CHAIN — STORE	1 : N
Employs	STORE — EMPLOYEE	1 : N
Supplies	SUPPLIER — PRODUCT	M : N
Stocks (identifying)	STORE — INVENTORY	1 : N
Tracked In	PRODUCT — INVENTORY	1 : N
Makes	CUSTOMER — SALES TRANSACTION	1 : N
Processes	CASHIER — SALES TRANSACTION	1 : N
Contains (Line Item)	SALES TRANSACTION — PRODUCT	M : N

The Complete EER Diagram — Retail Chain Management System



Key Takeaways

1

EER extends ER purposefully

Specialization, generalization, categories and constraints exist to model real hierarchies — not to add complexity for its own sake.

2

Two hierarchies drive the RCMS design

STORE and EMPLOYEE both split into disjoint, total subclasses — the defining EER feature of this schema.

3

Weak entities capture dependency

INVENTORY only exists in relation to a STORE, modelled correctly with a double outline and identifying relationship.

4

The diagram mirrors the business

Every symbol on the page — chain, store formats, staff roles, sales flow — traces back to a real operational rule.